

Deadlock (Urban / Close-Range Urban Warfare) Map Design

Executive Summary

Deadlock is a dense, war-torn urban battlefield designed for brutal close-quarters combat, rapid autonomous decision-making, and high-intensity chokepoint engagements. The map features collapsed buildings, tight alley networks, fortified intersections, underground metro tunnels, and rooftop traversal routes. It emphasizes Obsidian Protocol's autonomous tactics, sensor-limited environments, communications warfare, and logistics under urban constraint. In AR (Realtime/VR), the map anchors at ~4-5 m viewing distance, covering a 10-12 m physical area representing "1 km² of dense city terrain. Key features include multi-level combat zones, interior building routes, barricaded streets, and critical objective hubs.

Design Goals & Gameplay

Close-Range Urban Combat

Deadlock focuses on interior and tight-vehicle engagements. Streets are narrow, visibility is limited, and combat is fast. Buildings create vertically and interior routes that allow ambushes and flanking.

Autonomy & Micro-Flanking

Autonomous AI can detect ally threats, breach interior rooms, or repurpose for rooftops. Units adapt to collapsing structures, blocked roads, and shifting cover.

Sensor & Information Warfare

Units utilize sensor relay (e.g., thermal occlusion, and acoustic disruption). Rooftop relay towers restore sensor clarity. Destroying relay plunges entire districts into fog-of-war.

Logistics & Resources

Supply trucks must navigate blocked roads and delays. Control the **Central Transit Hub** and **Logistics Depot** enables motor resupply advances. Urban tactics are slow and vulnerable.

Deployment & Progress

A fast deployment budget (~10,000 points) forms players to choose between heavy breaching units or fast recon squads. Blue deploys from the southwest district, Red deploys from the northeast high-rise zone.

Spatial Constraints (Nreal AR)

Device Capabilities

Nreal LightAR: ~15° diagonal FOV, 1800x1080 per-eye resolution. Optimal viewing distance ~4-5 m.

Playable Area & Scaling

10x10 m physical area = 1 km² urban district (1,000 units). Designed for tabletop AR.

Tracking & Occlusion

Real-time tracking across horizontal planes. Depth-mach (if available) provides limited occlusion. Virtual objects render over real-world view, assume minimal real-world occlusion.

Layout Overview

Coordinate System

(0,0,0) is grid (0,0) area = 1 m. Origin (0,0) = southwest corner.

Key Locations

Location	Coordinate	Size	Description
Blue HQ B	0, 0	120x120+	Police station + fortified street barricade
Red HQ B	100, 100	120x120+	Corporate tower glass with corner road
Central Tn	500, 500	200x200+	Market entrances, bus terminals, collapsed platforms
Logistics D	300, 300	80x80 m	Supply trucks, crates, fuel drums
HighRise	700, 300	150x150+	Tall buildings, rooftop traversal routes
Old Market	200, 300	120x120+	Tight alleys, shops, interior routes
Underpass	400, 0+	—	Subterranean flanking route
Skyscraper	800, 700+	—	Barricaded skyline connecting rooftops
River & Bt	0, 450+	—	Two urban bridges at (100, 450) and (300, 450)
Observatory	900, 200	>100 m	Sensor vantage point
Neutral Pt	500, 900	—	Drone resupply zone

Chokepoints & Sightlines

Main Intersection (Deadlock Point)

The central intersection at (250, 250) is the narrative chokepoint — four streets converge, all partially blocked by debris.

Metro Tunnel

Low visibility, tight corridors, ideal for ambushes.

Skybridge Network

High-ground corridors with long sightlines but exposed positions.

Bridges

Critical for cross-district movement; controlling them prevents flanking.

Sightlines

Observation Tower provides long-range sensor coverage (~250 m). Alley networks create heavy fog-of-war pockets.

Deployment Zones

Blue Deployment: SW District + rear rally point.

Red Deployment: NE High-Rise Zone + rear rally point.

Neutral Forward Zone: Safehouse at (700, 900) for drone drops.

Assets & Environment

Terrain / Structures

Collapsed buildings, debris piles, broken asphalt, interior rooms, rooftops, skybridges.

Vegetation

Scattered urban trees, planter boxes, grass patches.

Buildings & Props

Urban Apartments, shops, offices, parking garages.

Military Barricades, sandbags, watchtowers.

Logistics Trucks, crates, fuel tanks, fuel drums.

Turret: Heavy anti-air, platforms, rails.

Effects

Smoke plumes, sparks, flickering lights, dust clouds.

Audio

Drums, distant explosions, radio chatter, wind between buildings.

LOD & Budget

Asset Cat

Count

Tx/Item

Total Size

Buildings

20

1.2M

Roads/Rails

6

10K

10K

Debris/Props

80

1K

40K

Vegetation

100

2K

20K

Transit Ave

20

20K

Total

—

~2.62M

Texture Budget

2048 for buildings/walls, 512x for props, 1024x for props. Target 1100 MB.

Unreal / Nreal Integration

Engine & Plugins

Unreal 5.4 (Lumen) + Nreal SDK (NREAL). ARKit/ARCore enabled.

Rendering

Forward rendering, stationary lights, baked GI, aggressive culling.

Markers

Single spatial anchor for entire map.

Interaction

Phone/glasses for panning/zooming; HUD on screen-space widgets.

Multitasking

Client-server architecture; offload AI physics to server.

Implementation Plan & Testing

Milestones

Milestone Timeline

1. Prototype Month 1: Block-out buildings, streets, metro, AI anchor test

2. Core Dev Month 2: Non-destructible, autonomous behaviors

3. AR Integration Month 3: Anchors, phone detection, basic interactivity

4. Content Dev Month 4: Final models, textures, lighting, audio

5. Gameplay Month 5: Objectives, balance, VFX (smoke, sparks)

6. Testing Month 6: Multiplayer stress test, sensor validation

Testing Plan

Autonomy

Verify AI reasoning logic, ally navigation, rooftop traversal.

Comms/Sensors

Test relay distribution, fog-of-war updates, LOD checks.

Performance

480 Hz AR render on target device.

QA Checklist

Navigation, cover usage, deployment budget enforcement, anchor drift, sensor accuracy.